

Capstone Project Document

AI-powered Film Photography Contest Management Platform	
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Finally, although we have worked hard to ensure accuracy, errors may still exist. We welcome any feedback and suggestions from our lecturers to further improve this project.

Definition and Acronyms

Acronym	Definition	Description
API	Application Programming Interface	A set of rules and protocols that allows different software applications to communicate with each other.
BA	Business Analysis	The practice of enabling change in an enterprise by defining needs and recommending solutions that deliver value to stakeholders.
BR	Business Rule	A rule that defines or constrains some aspect of business and always resolves to either true or false.
ERD	Entity Relationship Diagram	A flowchart that illustrates how "entities" such as people, objects, or concepts relate to each other within a system.

Acronym	Definition	Description
EXIF	Exchangeable Image File Format	A standard format for specifying meta-data tags embedded in digital image files (e.g., Camera Model, Lens, Film Stock/ISO).
GUI	Graphical User Interface	A type of user interface that allows users to interact with electronic devices through graphical icons and visual indicators.
MHs	Man-Hours	An estimated unit of work equal to one hour of continuous labor by one project team member.
pHash	Perceptual Hashing	A fingerprinting algorithm used to analyze image contents and generate a unique hash for duplicate image detection.
PM	Project Manager	The person responsible for planning, executing, monitoring, controlling, and closing projects.
RBAC	Role-Based Access Control	A security approach that restricts system access based on defined user roles (e.g., Candidate, Judge, Organizer).
SDD	Software Design Description	A document that describes the design of a software system, including its architecture, components, and interfaces.
SPMP	Software Project Management Plan	A document that defines the project plan, specifying the tasks, schedule, resources, and management approach.
SRS	Software Requirement Specification	A document that describes what the software will do and how it will be expected to perform.
UAT	User Acceptance Test	The final phase of the software testing process where actual software users test the software to make sure it can handle required tasks in real-world scenarios.
UC	Use Case	A list of actions or event steps typically defining the interactions between a role and a system to achieve a goal.

I. Project Introduction

1 Overview

1.1 Project Information

Project Name: AI-powered Film Photography Contest Management Platform

Software Type: Web Application

Purpose: To streamline the organization, submission, judging, and exhibition processes of film photography competitions.

Group Name: Group 7

2 Project Team

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3 Product Background

Film photography has become increasingly popular in recent years, leading to a growing number of photography contests. However, most contests are still managed using separate tools such as online forms, spreadsheets, emails, and cloud storage, resulting in inefficient workflows and difficulties in submission verification, judging, and data management.

Unlike digital photography contests, film photography requires additional technical metadata such as film stock, camera, lens, ISO, developing laboratory, and scanning information. Existing contest management systems generally do not support these specialized requirements.

To address these limitations, this project proposes an AI-powered Film Photography Contest Management Platform. The platform centralizes contest management, supports film-specific metadata, and integrates AI for duplicate image detection, AI-generated image detection, automatic image categorization, and contest analytics, improving efficiency, transparency, and fairness throughout the competition process.

4 Existing Systems

4.1 Picter

Link: <https://www.picter.com>

Description: A professional platform for managing photography competitions, open calls, grants, and awards. It is widely used by organizations such as World Press Photo, Leica, and Magnum.

Actors: Organizers, judges, photographers, administrators.

Key Features:

- Online contest creation and management.
- Photo submission and portfolio management.
- Online judging and scoring.
- Result publication and digital archive.

Pros: Complete end-to-end contest management; professional judging workflow; supports large-scale international competitions.

Cons: Not designed specifically for film photography; does not provide film-specific metadata management (film stock, camera, developing lab, etc.); limited AI verification features.

4.2 CameraComp

Link: <https://www.cameracomp.com>

Description: A cloud-based competition management system developed for photography clubs and federations.

Actors: Club administrators, judges, photographers.

Key Features:

- Competition management.
- Image submission.
- Online and offline judging.
- Member and event management.

Pros: Simple and intuitive interface; integrated competition and member management; supports multiple judging methods.

Cons: Focuses on photography clubs rather than film photography contests; no AI-generated image detection; no film metadata management.

Compared with the systems above, the proposed platform is specifically designed for film photography contests. It focuses on film-specific metadata, AI-based verification, and a more specialized workflow for organizers, judges, and participants.

5 Business Opportunity

The growing popularity of film photography has increased the demand for efficient contest management. However, most organizers still use separate tools, making submission management, judging, and verification inefficient. This project provides a centralized platform with AI support to improve efficiency, transparency, and fairness for photography contests.

6 Software Product Vision

The project aims to develop a web and mobile platform for managing film photography contests. It supports the complete contest workflow, from registration and submission to judging and result publication, while integrating AI for duplicate image detection, AI-generated image detection, and automated image categorization to enhance contest management.

7 Project Scope & Limitations

7.1 Major Features

The following primary functional modules are included:

- **FE1 – User & Role Management:** Provides authentication, role-based access control (RBAC), and profile management for Candidates, Judges, and Organizers/Admins.
- **FE2 – Contest Management:** Enables Organizers to create and configure new contests, set submission rules, manage timelines, oversee participant lists, and assign judges.
- **FE3 – Submission System:** Allows candidates to upload high-resolution film scan images, input detailed technical specifications (EXIF metadata, Camera, Lens, Film Stock/ISO, Aperture, Shutter Speed), and track submission statuses.
- **FE4 – Judging & Evaluation:** Provides a dedicated judging panel featuring high-resolution image viewing tools (Zoom/Pan), multi-criteria evaluation scoring, and feedback/commenting capabilities.

- **FE5 – Automated Processing & Anti-cheat:** Automatically extracts EXIF data from uploaded images and utilizes perceptual hashing (pHash) with Hamming distance calculations to detect and flag potential image plagiarism or duplicate submissions.
- **FE6 – Exhibition Gallery:** A public online gallery showcasing winning or selected submissions, equipped with advanced filtering options by Film Stock, Camera model, or Contest.

7.2 Limitations & Exclusions

The following points detail the operational constraints and out-of-scope boundaries:

- **LI-01:** File size and format restrictions: Supports only .jpg, .jpeg, and .png formats, with file sizes strictly limited between 2 MB and 25 MB.
- **LI-02:** Automated metadata extraction depends entirely on intact EXIF data within uploaded film scan files; stripped EXIF files require manual input and administrative flagging.
- **LI-03:** Perceptual hashing (pHash) and Hamming distance duplication detection are processed asynchronously via message queues, resulting in slight processing delays rather than real-time feedback upon upload.
- **LI-04:** No support for native mobile applications (iOS/Android); the system is strictly developed as a responsive web application.
- **LI-05:** No integration with online payment gateways or photo editing/post-processing tools (candidates must upload finalized image files, and all financial transactions are out of scope).

II. Project Management Plan

1 Overview

1.1 Scope & Estimation

WBS	Item	Complexity	Est. Effort (MHs)
1	FE01. User & Role Management		40
1.1	Authentication System (Login, Register, OAuth Google, JWT)	Simple	10
1.2	Password Recovery & Email Verification	Simple	8
1.3	User Profile & Portfolio Management	Medium	10
1.4	Role-Based Access Control (RBAC) & User Management (Admin)	Medium	12
2	FE02. Contest Management		48
2.1	Contest Creation & Basic Setup (Rules, Prizes, Categories)	Medium	14
2.2	Timeline & Phase Transition Configuration (Auto Scheduler)	Medium	12
2.3	Jury Assignment & Judging Criteria Setup	Medium	12
2.4	Contest Status Management & Moderator Dashboard	Simple	10
3	FE03. Submission System		64
3.1	Presigned Cloud Storage Upload (S3/Cloudinary) for Hi-Res Scan	Complex	20
3.2	Film Specifications & Metadata Input Form (Film Stock, Camera, Lens)	Medium	14
3.3	Negative Film / Contact Sheet Proof Attachment Upload	Medium	16
3.4	Submission Drafts & Candidate Entry History	Simple	14
4	FE04. Judging & Evaluation		56
4.1	Blind Judging Portal (Anonymized Candidate Data)	Medium	12
4.2	High-Resolution Image Viewer (Zoom/Pan HD & Negative Inspector)	Medium	16
4.3	Multi-Criteria Scoring System & Feedback Panel	Complex	16
4.4	Final Leaderboard Calculation & Winner Approval	Medium	12
5	FE05. Automated Processing & Anti-cheat (AI Subsystem)		72
5.1	Auto EXIF Metadata Extraction Engine	Medium	18

WBS	Item	Complexity	Est. Effort (MHs)
5.2	EXIF vs User Declaration Mismatch Verification	Medium	14
5.3	pHash Duplication Detection & Fingerprinting Algorithm	Complex	24
5.4	Fraud Flagging System & Admin Inspection Dashboard	Medium	16
6	FE06. Exhibition Gallery & Community		32
6.1	Public Gallery & Masonry Display Grid	Simple	8
6.2	Advanced Filtering (Film Stock, Camera Model, Contest, Year)	Medium	10
6.3	Public Voting & Anti-Spam Community Engagement	Medium	8
6.4	Social Sharing & Certificate Generation for Winners	Simple	6
7	FE07. System Architecture, QA & DevOps		88
7.1	SRS / SDD Documentation & System Architecture Design	Medium	24
7.2	Database Schema Design (ERD) & Base API Setup	Medium	16
7.3	QA Test Plan, Test Cases Execution & Bug Fixing	Medium	30
7.4	Docker Containerization, CI/CD Pipeline & Deployment	Medium	18
TOTAL			400 MHs

1.2 Project Objectives

Testing Stage	Test Coverage	No. of Defects	% of Defect	Notes
Reviewing	100% artifacts reviewed (SRS, SDD, ERD, Wireframes)	-	-	Peer review and supervisor feedback
Unit Test	$\geq 70\%$ code coverage for Core BE and AI modules	0 critical; ≤ 10 minor	$\leq 10\%$	Automated testing (JUnit, Jest, PyTest for AI functions)
Integration Test	$\geq 85\%$ API and async pipeline coverage	0 critical; ≤ 8 minor	$\leq 8\%$	Postman / Swagger and S3 upload / AI queue testing
System Test	$\geq 95\%$ major features (Contest, Submission, AI Inspection, Judging)	0 critical; ≤ 5 minor	$\leq 5\%$	Staging environment regression and performance testing for high-resolution uploads
Acceptance Test (UAT)	100% critical user flows and AI verification accuracy	0 critical / blocker	0%	UAT with stakeholders (organizers, jury, candidates)

1.3 Project Risks

No.	Risk Description	Impact	Possibility	Response Plans
1	Changing requirements from stakeholders (organizers, jury)	Medium	Medium	Baseline SRS plus change control process; weekly review and sign-off with supervisor
2	Limited domain knowledge in film photography (film stocks, EXIF structures, scan types) and AI anti-cheat models	High	Medium	Consult film photography experts/advisors; early PoC for EXIF extraction and pHash algorithm; prototype feedback
3	Third-party integrations and services (AWS S3 / Cloudinary for high-resolution image storage, Docker, Redis, Resend / SMTP) are unstable or exceed free-tier limits	Medium	Medium	Early PoC in sandbox; image compression and thumbnail generation; retries and fallback mechanisms; service cost monitoring
4	Data privacy and security (protecting unpublished contest entries and candidate PII)	High	Low	Implement RBAC, anonymized submission pipeline for blind judging, S3 presigned URLs with expiration, and data encryption at rest and in transit
5	Resource availability or absence among team members	Medium	Medium	Cross-training tech stacks (front-end / back-end / AI); backup feature ownership; daily standups; sprint buffer time
6	Performance degradation under peak loads (simultaneous high-resolution image uploads near contest deadline)	High	Medium	Direct-to-S3 presigned upload (bypass core server); Redis caching; asynchronous background AI processing queue; rate limiting and load testing in Sprint 6

2 Management Approach

2.1 Project Process

Methodology: The project applies the Agile Scrum methodology.

Iteration Model: 1-week sprints, total of 7 sprints (7 weeks).

Sprint Breakdown:

- **Sprint 1:** Project initiation, requirement elicitation (SRS), architecture & DB design (SDD), UI/UX wireframing, and AI Proof of Concept (EXIF & pHash).
- **Sprint 2:** Development of FE01 (User Authentication, Profile Management, & RBAC).
- **Sprint 3:** Development of FE02 (Contest Creation, Timeline, & Jury Assignment).
- **Sprint 4:** Development of FE03 (Submission System with Presigned S3 Upload & Film Specifications).
- **Sprint 5:** Development of FE04 (Blind Judging Portal, HD Image Viewer, & Scoring System).
- **Sprint 6:** Development of FE05 (AI Processing Pipeline: Auto EXIF & pHash Duplicate Detection) & FE06 (Exhibition Gallery).
- **Sprint 7:** FE07 Integration, System Testing, Bug Fixing, UAT, Docker Containerization, & Cloud Deployment.

Scrum Ceremonies: Daily Stand-up (15 minutes), Weekly Sprint Planning, Sprint Review, and Sprint Retrospective.

Artifacts: Product Backlog, Sprint Backlog, Burndown Chart, Weekly Working Increment.

2.2 Quality Management

The quality objective is to ensure a stable, accurate, secure, and performant platform for managing film photography contests and automated image inspection within the 7-week timeframe.

The approach includes:

- **Defect Prevention:**
 - Enforce clean code conventions and modular architecture (Separation of Core API and AI Services).
 - Database normalization (PostgreSQL), constraints, and indexing for fast query performance on media records.
- **Reviewing:**

- Mandatory Code Review with at least 1 Peer Reviewer and Leader approval before merging Pull Requests.
- Peer review for technical documentation (SRS, SDD, ERD, Test Cases).

- **Unit Testing:**

- **Tools:** Jest/React Testing Library (Frontend), JUnit/PyTest (Core BE & AI Engine).
- **Goal:** $\geq 70\%$ code coverage, focusing on critical API logic, EXIF parsers, and hashing algorithm utilities.

- **Integration Testing:**

- **Tools:** Postman / Newman for REST API workflows.
- **Scope:** Asynchronous task queue handling (Redis/Celery) for background image processing and Direct-to-Cloud (AWS S3 / Cloudinary) presigned upload pipelines.

- **System Testing:**

- Conducted in Week 7 on a Staging Environment hosted on AWS / Cloud VPS.
- Full regression testing of all modules (FE01 to FE07).
- Performance and Load Testing (using Apache JMeter or K6) for peak submission scenarios near contest deadlines and Hi-Res image viewing.

- **Acceptance Testing (UAT):**

- User Acceptance Testing (Week 7) with primary stakeholders: Contest Organizers, Jury Members, and Candidates/Photographers.
- Ensure 100% of critical flows (Contest creation, Entry submission, AI flagging, and Blind judging) pass successfully before final presentation.

2.3 Training Plan

Training Area	Participants	When, Duration	Waiver Criteria
Next.js / ReactJS (Frontend)	Uyen, Tan	Week 1, 2 days	≥ 1 year React experience
Node.js / Java Spring (Backend)	Khanh, Tai	Week 1, 2 days	≥ 1 year Backend experience

Python & OpenCV / pHash / EXIF Parsing	Thien	Week 1, 2 days	Prior AI / Image Processing experience
PostgreSQL Schema & Media Indexing	All	Week 1, 1 day	Prior DB design experience
Git/GitHub Workflow (PR, Code Review)	All	Week 1, 0.5 day	Familiar with Git flow
AWS S3 / Cloudinary Presigned Direct Upload	Khanh, Uyen, Tan	Week 2, 1 day	Prior Cloud Storage experience
Security & RBAC Authorization (JWT)	All	Week 2, 0.5 day	Completed Security course
Redis Queue & Async Task Handling	Khanh, Thien	Week 3, 1 day	Prior Message Queue experience
Deep Zoom HD Image Viewer (OpenSeadragon)	Uyen, Tan	Week 4, 0.5 day	Prior Canvas/Image lib experience
Docker Containerization & Cloud Deployment	Tai, Khanh	Week 6, 1 day	Prior DevOps/Docker experience

3 Project Deliverables

No.	Deliverable	Due Date	Notes
1	Project Charter, SRS, SDD & AI PoC Report	Sprint 1 (Week 1)	
2	Authentication & User Management Module (FE01)	Sprint 2 (Week 2)	
3	Contest Management & Jury Assignment Module (FE02)	Sprint 3 (Week 3)	
4	Submission System & Direct Cloud Storage (FE03)	Sprint 4 (Week 4)	
5	Blind Judging Portal & HD Scoring System (FE04)	Sprint 5 (Week 5)	
6	AI Pipeline Integration & Exhibition Gallery (FE05 & FE06)	Sprint 6 (Week 6)	
7	Integration Test, Docker Package & Cloud Deployment (FE07)	Sprint 7 (Week 7)	

4 Responsibility Assignments

Responsibility	Tai	Thien	Khanh	Uyen	Tan
Project Planning & Tracking	D	R	S	R	I
Prepare Project Charter & SRS Document	D	R	S	R	S
Architecture & DB Design (SDD, ERD)	S	R	D	I	I
UI/UX Wireframing & Design (Figma)	R	I	I	D	D
Core Backend API Implementation (Spring Boot)	S	I	D	I	I
Direct Cloud Storage & Presigned URL Integration	I	S	D	I	R
AI Pipeline Implementation (EXIF & pHash Engine)	R	D	S	I	I
Frontend Implementation - Candidate & Gallery (FE03, FE06)	I	I	I	S	D
Frontend Implementation - Jury Portal (FE04)	I	I	I	D	S
System Integration & Testing (Testing & QA)	R	R	R	R	D
Final Deployment & Project Documentation	D	R	R	R	R

5 Project Communications

Communication Item	Who/ Target	Purpose	When, Frequency	Type, Tool, Method(s)
Weekly Planning Meeting	All team members	Plan tasks for upcoming sprint, review SRS/SDD, and assign roles	Every Friday (1 session/week)	Online meeting (Google Meet)
Weekly Progress Review	All team members	Share development progress, adjust timeline, discuss blockers	Every Wednesday (1 session/week)	Google Docs, Zalo group

Internal Feature Demo	Relevant developers + Reviewer (Leader)	Validate completed modules (Spring Boot, Python AI, React UI) before merging	After each feature completion	Live screen share via Google Meet or Zalo video
Documentation Review	Assigned members (Tai, Khanh)	Review and update SRS, SDD, ERD, and User Manual documents	Every week	Google Docs
Code Review Session	Feature developer + Reviewer	Review code logic, API endpoints, pHash/EXIF algorithms, and coding conventions	Before merging Pull Request to main branch	GitHub Pull Request
Urgent Bug or Blocker Sync	Involved members only	Quickly resolve critical technical issues, API mismatch, or Cloud upload errors	Ad-hoc (as needed)	Zalo Chat / Call
Shared Asset Management	All members	Store and collaborate on design files (Figma), diagrams (Draw.io), docs, and slide presentations	Continuous	Google Drive / Figma / GitHub

6 Configuration Management & Infrastructure

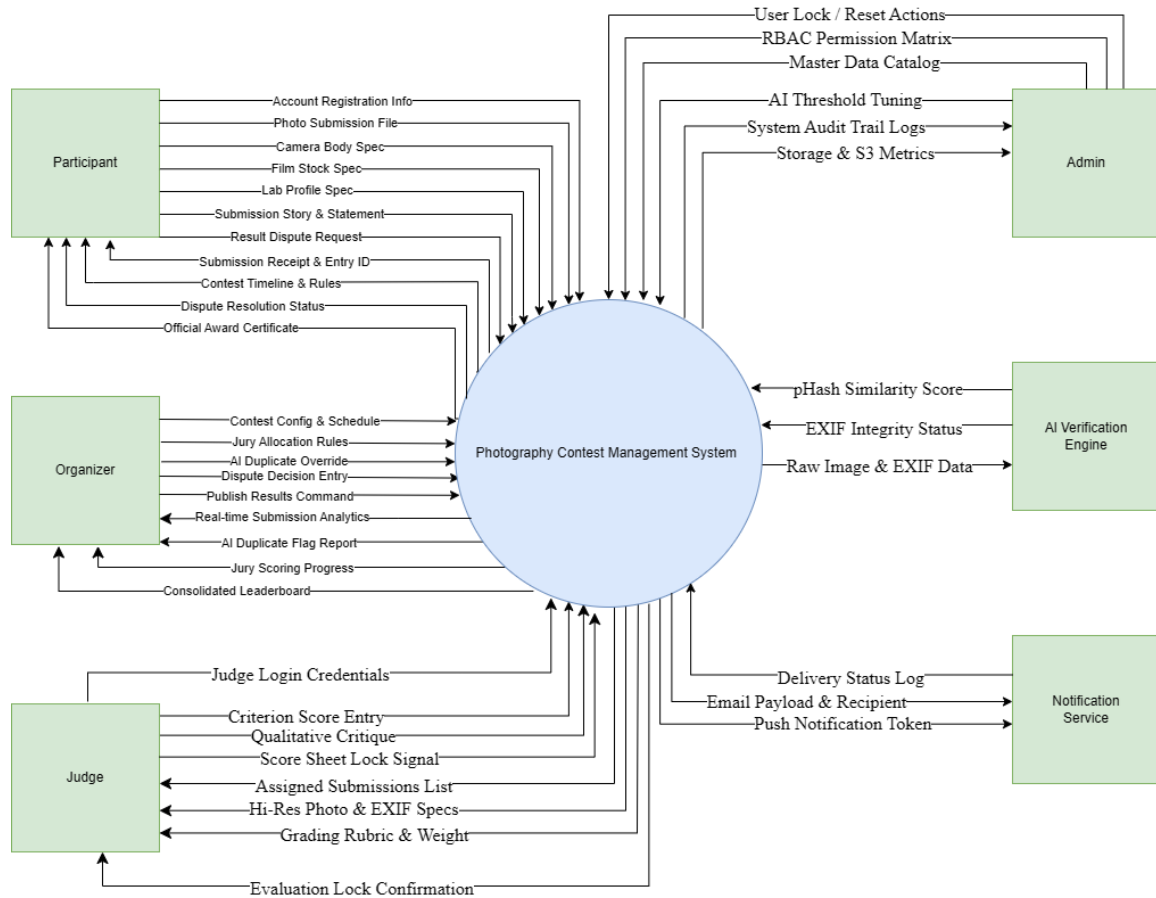
Category	Tools / Infrastructure
Frontend	React / Next.js
Backend	Java Spring Boot, Python FastAPI (AI Subsystem)
Database & Cache	PostgreSQL
Cloud Storage	AWS S3 / Cloudinary (Presigned / Signed URL)
IDEs / Editors	VS Code, PyCharm
Diagramming & UI	Figma, Draw.io
Documentation & VCS	Google Docs, GitHub
Testing & API Spec	Postman, Swagger / OpenAPI

III. Software Requirement Specification

1 Product Overview

The AI-powered Film Photography Contest Management Platform is a web-based application designed to manage and automate the entire workflow of a film photography competition. The system allows participants to register, submit photographs, and track the status of their submissions. Organizers can create and manage contests, assign judges, review flagged submissions, and publish final results. Judges evaluate photographs based on predefined criteria, while administrators manage user accounts, permissions, and system configurations.

To improve efficiency, fairness, and submission authenticity, the platform integrates an AI Verification Engine to extract EXIF metadata, detect duplicate images, and identify AI-generated content before submissions proceed to the judging stage. A Notification Service delivers important updates, including registration confirmations, submission status, judging progress, and contest results. Overall, the system provides a transparent, secure, and efficient environment for managing film photography competitions.



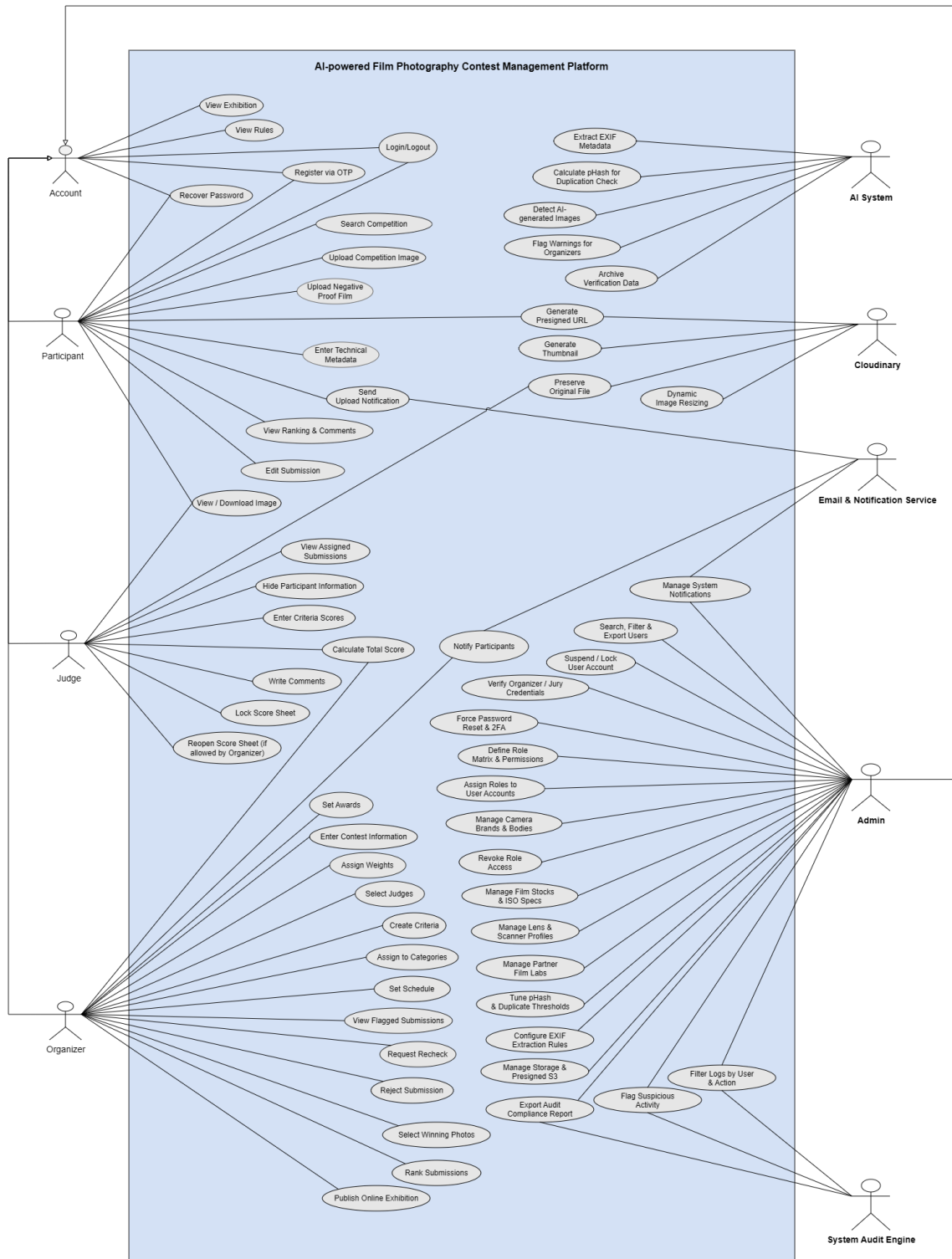
2 User Requirements

2.1 System Actors

No.	Actor	Description
1	Administrator	Responsible for overall system configuration, managing user accounts, enforcing RBAC (Role-Based Access Control) permissions, tuning AI detection thresholds, managing camera/film master data catalogs, and monitoring system audit trail logs.
2	Organizer	A contest management authority responsible for setting up competitions, configuring evaluation rubrics, allocating judges, reviewing AI duplicate/fraud flags, handling dispute entries, and publishing final competition results.
3	Judge	A designated evaluator responsible for logging in, viewing assigned submission queues, evaluating submissions against multi-criteria rubrics, providing qualitative feedback, and locking score sheets.
4	Participant	A registered user (candidate) who uses the system to browse contests, submit photo entries along with camera/film metadata, track submission status, receive evaluation feedback, and submit result dispute requests.
5	AI Verification Engine (System Actor)	An external automated service that processes uploaded raw images and EXIF data to compute pHash similarity scores, detect potential duplicate submissions, and flag AI-generated or manipulated images.
6	Notification Service (System Actor)	An external communication provider responsible for delivering transaction emails, submission receipts, evaluation status updates, and system push notifications to participants and staff.

2.2 Use Cases

2.2.1 Use Case Diagram



2.2.2 Use Case Descriptions

ID	Use Case	Actors	Use Case Description
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UC-01	View Exhibition	Account	Allows users to view the exhibition gallery displaying award-winning photo entries from past contests.
UC-02	View Rules	Account	Displays the rules and regulations of a specific photography contest for users to review before participating.
UC-03	Login/Logout	Account, Participant	Authenticates users via account credentials to grant access to the system, and allows secure session termination via Logout.
UC-04	Register via OTP	Account, Participant	Allows users to register a new account using OTP (One-Time Password) verification sent to their email.
UC-05	Recover Password	Account, Participant	Allows users to reset their forgotten password through email verification.
UC-06	Search Competition	Participant	Allows participants to search and filter photography competitions by name, category, or status.
UC-07	Upload Competition Images	Participant	Allows participants to upload their photo entries for a specific competition.
UC-08	Upload Negative Proof Film	Participant	Allows participants to upload negative proof film images as evidence to verify the authenticity of their photo entries.
UC-09	Enter Technical Metadata	Participant	Allows participants to input technical metadata for their submitted photo, such as camera, lens, ISO, aperture, and shutter speed.
UC-10	Send Upload Notification	Participant, Email & Notification Service	Automatically sends a notification to the participant's email confirming that their upload has been received.
UC-11	Generate Presigned URL	Participant, Cloudinary	Generates a secure, time-limited URL that allows the participant's browser to upload images directly to Cloudinary storage.
UC-12	View Ranking & Comments	Participant	Allows participants to view their ranking results and comments left by judges for their submitted entries.

UC-13	Edit Submission	Participant	Allows participants to edit the details of a previously submitted entry before the submission deadline.
UC-14	View/Download Image	Participant, Judge	Allows participants and judges to view or download submitted images in their original quality for review or reference.
UC-15	Preserve Original File	Judge, Cloudinary	Ensures that the original uploaded image is securely stored in Cloudinary without modification, allowing judges to access the original file whenever needed.
UC-16	View Assigned Submissions	Judge	Allows the Judge to view the list of submissions assigned to them by the Organizer for a given contest category or judging round, including entry thumbnail, ID, and technical metadata, before starting evaluation.
UC-17	Hide Participant Information	Judge	Enables blind (anonymous) judging by hiding the submitting participant's identifying information (name, account, portfolio link) from the Judge's view during scoring, ensuring an unbiased evaluation.
UC-18	Enter Criteria Scores	Judge	Allows the Judge to input a numerical score for each submission against every evaluation criterion (e.g., creativity, composition, technical execution, film grain) defined by the Organizer for that contest.
UC-19	Calculate Total Score	Judge, Organizer	Automatically computes the weighted total score for each submission by combining the individual criterion scores entered by the Judge with the weight assigned to each criterion, used by the Organizer to determine rankings.
UC-20	Write Comments	Judge	Allows the Judge to add a qualitative critique or feedback text for a submission alongside its numeric score, providing context for the participant and the Organizer.

UC-21	Lock Score Sheet	Judge	Allows the Judge to lock their score sheet for a submission once evaluation is complete, preventing further edits and confirming the score as final.
UC-22	Reopen Score Sheet	Judge, Organizer	Allows the Organizer to unlock a previously locked score sheet (e.g., on the Judge's request or in case of an error/dispute), enabling the Judge to revise scores or comments before results are finalized.
UC-23	Flag Warmings for Organizers	Organizer, AI System	The AI System automatically flags a submission with a warning — such as a duplicate (pHash) match, suspected AI-generated content, or EXIF inconsistency — and sends the flag report to the Organizer for manual review.
UC-24	Notify Participants	Organizer, Email & Notification Service	Allows the Organizer to trigger notifications (email/push) to participants regarding contest updates, submission status, dispute outcomes, or final results, delivered through the Notification Service.
UC-25	Set Awards	Organizer	Allows the Organizer to define the award structure for a contest, including award categories, number of winners per category, and prize details.
UC-26	Enter Contest Information	Organizer	Allows the Organizer to create and configure a new contest by entering its name, theme, categories, timeline, and rules.
UC-27	Assign Weights	Organizer	Allows the Organizer to configure the weight (percentage contribution) of each judging criterion used when calculating a submission's total score.
UC-28	Select Judges	Organizer	Allows the Organizer to select and assign Judges to a contest or specific category, including allocating them to particular judging rounds.

UC-29	Create Criteria	Organizer	Allows the Organizer to define and manage the set of scoring criteria (e.g., creativity, storytelling, technical execution) that Judges will use to evaluate submissions in a contest.
UC-30	Assign to Categories	Organizer	Allows the Organizer to classify and assign submitted photographs to their appropriate contest categories or divisions for organized judging and ranking.
UC-31	Set Schedule	Organizer	Allows the Organizer to configure the contest timeline, including submission period, judging period, recheck deadline, and winner announcement schedule.
UC-32	View Fagged Submissions	Organizer	Displays the list of submissions flagged by the AI verification system for duplicate detection, AI-generated content, or metadata inconsistencies, allowing the Organizer to review them.
UC-33	Request Recheck	Organizer	Enables the Organizer to request an additional review for a submission when suspicious content or scoring discrepancies are identified.
UC-34	Reject Submission	Organizer	Allows the Organizer to reject submissions that violate contest regulations, fail verification, or do not meet eligibility requirements.
UC-35	Select Winning Photos	Organizer	Enables the Organizer to select winning submissions based on judges' scores, contest rules, and verification results before publication.
UC-36	Rank Submissions	Organizer	Automatically or manually ranks submissions according to judges' scores and predefined contest criteria to determine final standings.
UC-37	Publish Online Exhibition	Organizer	Allows the Organizer to publish approved winning photos to the online exhibition, making them available for public viewing.

UC-38	Extract EXIF Metadata	AI System	Automatically extracts EXIF metadata from uploaded images to verify camera information, capture settings, and image authenticity.
UC-39	Calculate pHash for Duplication Check	AI System	Generates perceptual hash (pHash) values to compare uploaded images and identify duplicate or highly similar submissions.
UC-40	Detect AI-generated Images	AI System	Analyzes uploaded images using AI detection techniques to identify whether the image was generated or manipulated by artificial intelligence.
UC-41	Archive Verification Data	AI System	Stores verification results, extracted metadata, and AI analysis records for auditing, dispute resolution, and future reference.
UC-42	Generate Thumbnail	Cloudinary	Automatically generates optimized thumbnail versions of uploaded images for previews, galleries, and faster loading.
UC-43	Dynamic Image Resizing	Cloudinary	Dynamically resizes images into different resolutions while preserving image quality for various display devices.
UC-44	Manage System Notifications	Admin, Email & Notification Service	Allows the Admin to configure notification templates and events, while the Email Notification Service delivers OTPs, status updates, and contest notifications to users.
UC-45	Search, Filter & Export Users	Admin	Enables the Admin to search, filter, and export user information for administration, reporting, and system management purposes.
UC-46	Suspend/Lock User Account	Admin	Allows the Admin to temporarily suspend or permanently lock user accounts upon detecting policy violations.
UC-47	Verify Organizer/Jury Credentials	Admin	Allows the Admin to verify profiles and approve the identities and credentials of Organizers or Jury members.
UC-48	Force Password Reset & 2FA	Admin	Allows the Admin to force users to reset their passwords and set up Two-Factor Authentication (2FA) upon their next login.

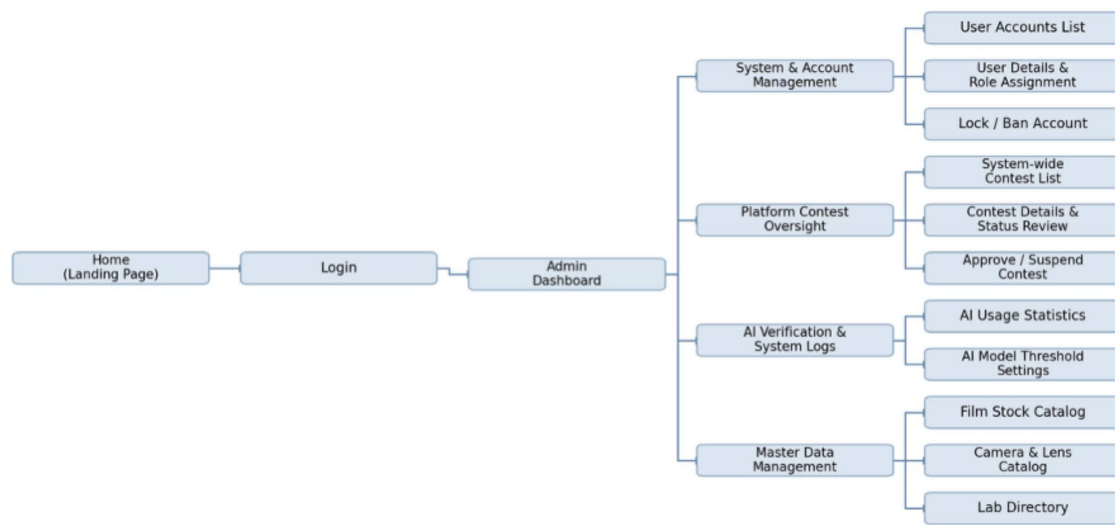
UC-49	Define Role Matrix & Permissions	Admin	Allows the Admin to configure the role-based access control (RBAC) matrix, specifying detailed permissions for each system role.
UC-50	Assign Roles to User Accounts	Admin	Allows the Admin to assign specific roles and privileges to individual user accounts within the system.
UC-51	Manage Camera Brands & Bodies	Admin	Allows the Admin to add, update, or delete categories of analog camera brands and camera bodies.
UC-52	Revoke Role Access	Admin	Allows the Admin to partially or fully revoke access permissions previously granted to a user account.
UC-53	Manage Film Stocks & ISO Specs	Admin	Allows the Admin to manage the catalog of film stocks and their corresponding ISO specifications.
UC-54	Manage Lens & Scanner Profiles	Admin	Allows the Admin to manage configurations and metadata profiles for various camera lenses and film scanners.
UC-55	Manage Partner Film Labs	Admin	Allows the Admin to add, update, or remove information regarding partner film processing labs.
UC-56	Tune pHash & Duplicate Thresholds	Admin	Allows the Admin to adjust the pHash algorithm and similarity thresholds to optimize the detection of duplicate or plagiarized images.
UC-57	Configure EXIF Extraction Rules	Admin	Allows the Admin to define rules for parsing and extracting EXIF metadata from uploaded images.
UC-58	Manage Storage & Presigned S3	Admin	Allows the Admin to manage storage capacity and configure secure upload links (Presigned URLs) via Amazon S3.
UC-59	Export Audit Compliance Report	Admin, System Audit Engine	Allows the generation and exporting of audit reports regarding system security compliance and data integrity.
UC-60	Flag Suspicious Activity	Admin, System Audit Engine	The system automatically flags suspicious activities (e.g., spam, hacking attempts) and alerts the Admin for resolution.

UC-61	Filter Logs by User	Admin, System Audit Engine	Allows the Admin to search, filter, and view detailed system activity logs for specific user accounts.
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3 Functional Requirements

3.1 System Functional Overview

3.1.1 Web Application Admin Screen Flow



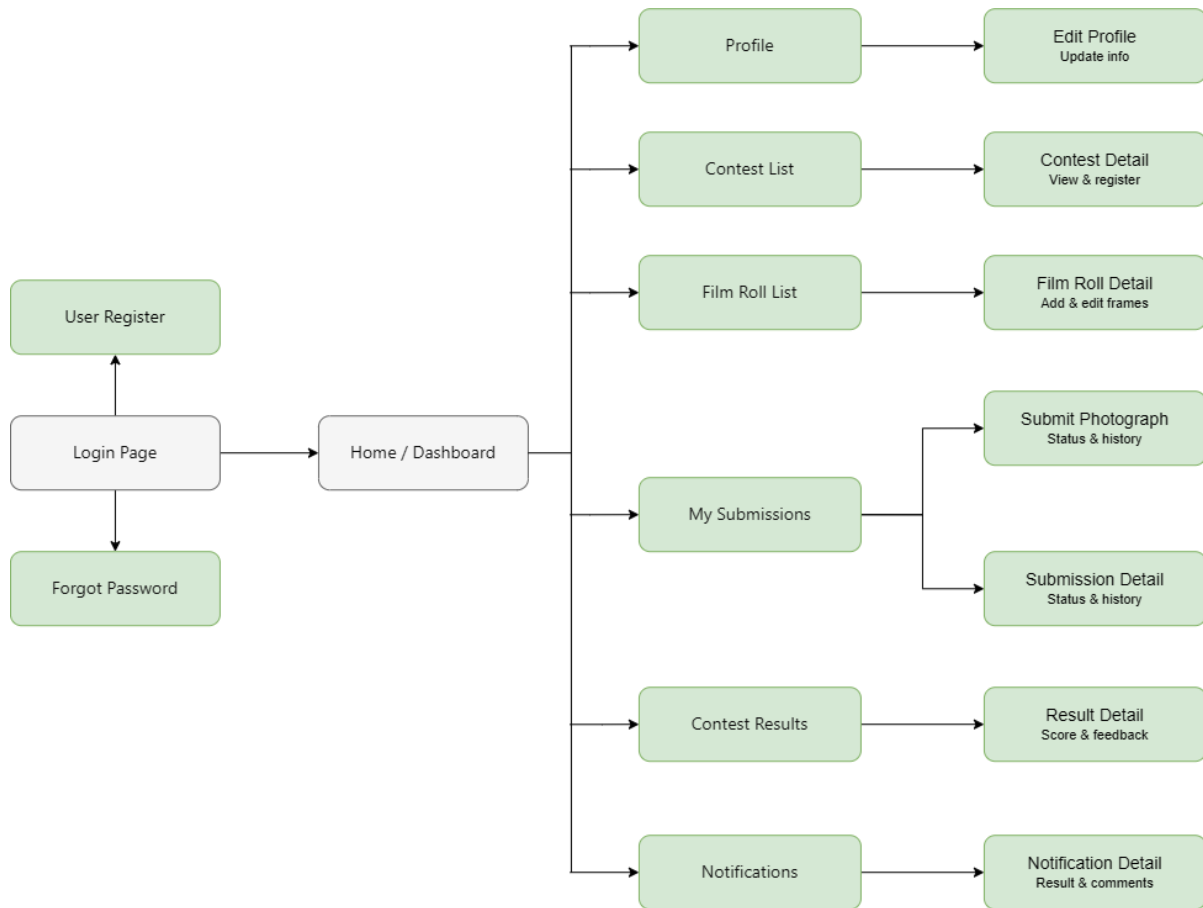
Admin's Screen Flow

3.1.2 Web Application Admin Screen Description

#	Feature	Screen	Description
1	General	Home	The public landing page of the system. Allows the Admin to view the general interface and quickly access the admin Login button.
2	General	Login	Authentication screen for the Admin account using Email/Password. After successful authentication, the system redirects directly to the Admin Dashboard.

#	Feature	Screen	Description
3	Dashboard	Admin Dashboard	The central screen displaying platform-wide KPIs: total number of contests, total submissions, and number of AI scans performed.
4	Account Management	User Accounts List	Manages the list of all user accounts on the system (Organizer, Judge, Participant).
5	Account Management	User Details & Role Assignment	View detailed account information and assign roles/permissions to the account.
6	Account Management	Lock / Ban Account	Screen for locking or unlocking a violating account.
7	Contest Oversight	System-wide Contest List	Lists all contests on the system for the Admin to oversee.
8	Contest Oversight	Contest Details & Status Review	View detailed content, rules, and progress of a specific contest.
9	Contest Oversight	Approve / Suspend Contest	Screen for approving a new contest or suspending a contest that violates policy.
10	AI & System Logs	AI Usage Statistics	Statistical report on the frequency and activity log of the AI-Generated Image Detection and Duplicate Detection models.
11	AI & System Logs	AI Model Threshold Settings	Screen for configuring the AI confidence threshold used to automatically flag violations.
12	Master Data	Film Stock Catalog	Catalog for managing film stock types (Kodak, Fuji, Ilford, etc.).
13	Master Data	Camera & Lens Catalog	Catalog for managing cameras and lenses.
14	Master Data	Lab Directory	List of managed, affiliated film-developing labs.

3.1.3 Web Application Participant Screen Flow

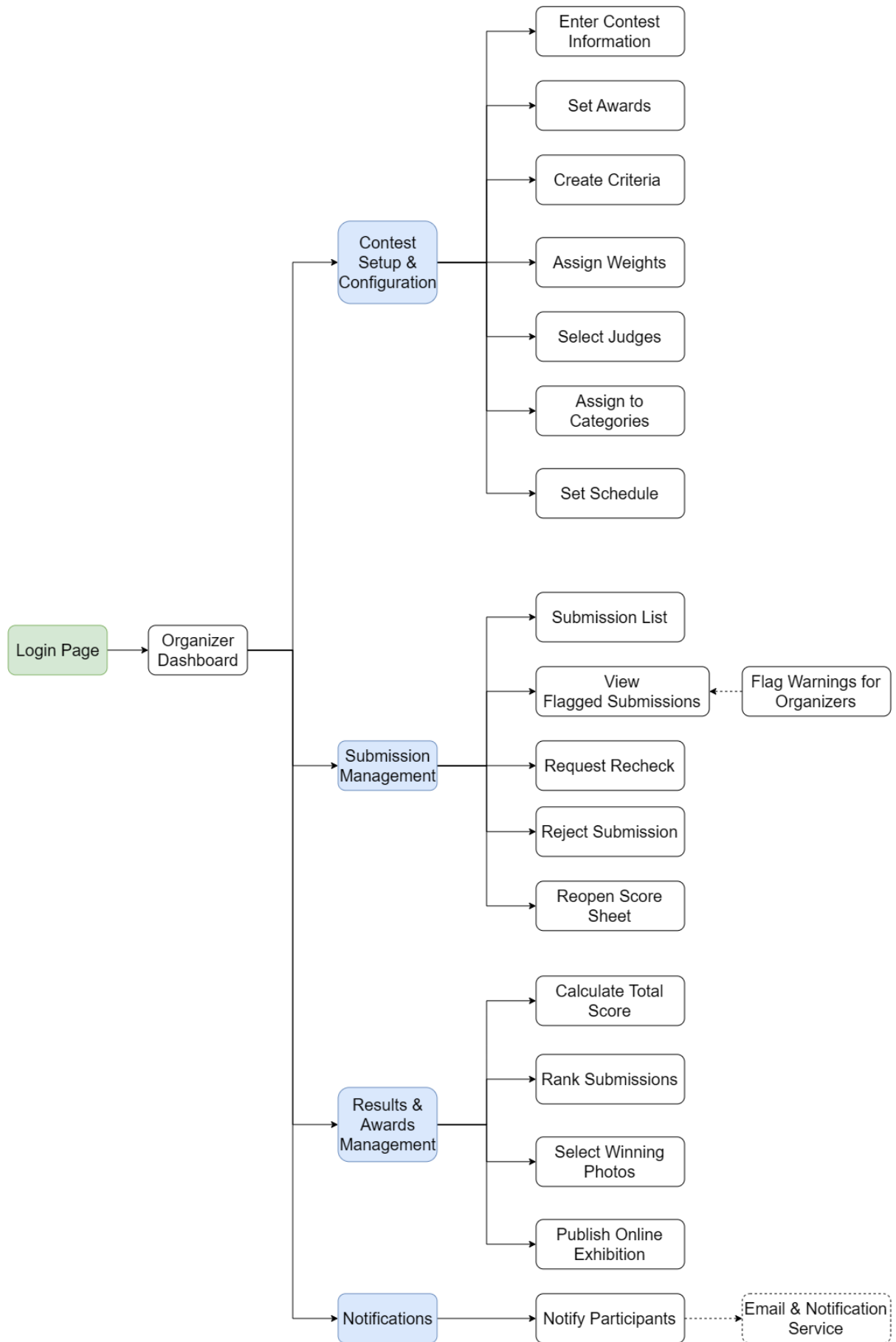


3.1.4 Web Application Participant Screen Description

#	Feature	Screen	Description
1	Authentication	Login Page	Allows the Participant (Film Photographer) to log in to the system with a registered account.
2	Register	User Register	Allows the Participant to create a new account to join the system.
3	Reset Password	Forgot Password	Allows the Participant to reset their password if forgotten.
4	Home Navigation	Home / Dashboard	Displays the main navigation to Profile, Contests, Film Rolls, Submissions, Results, and Notifications.
5	View Profile	Profile	Displays the Participant's personal profile information.

#	Feature	Screen	Description
6	Update Profile	Edit Profile	Allows the Participant to edit their personal information.
7	Contest Discovery	Contest List	Displays the list of film photography contests currently open for registration.
8	Contest Registration	Contest Detail	Displays contest details (theme, rules, deadline) and allows the Participant to register to participate.
9	Film Roll Management	Film Roll List	Displays the list of film rolls the Participant has recorded.
10	Film Roll & Frame Detail	Film Roll Detail	Allows the Participant to create a new film roll, view, add, and edit information for each frame (film stock, ISO, camera, lens, lab, etc.).
11	My Submissions	My Submissions	Displays the list of entries the Participant has submitted.
12	Submit Photograph	Submit Photograph	Allows the Participant to upload a scanned photo with technical metadata and select the related frame.
13	Submit Photograph	Submission Detail	Displays the processing status of the submission (submitted, under verification, approved, etc.).
14	Result Publication	Contest Results	Displays the results list and ranking of the contest.
15	Judging Feedback	Result Detail	Displays the score, ranking, and judge's feedback for each submission.
16	Notifications	Notifications	Displays the list of notifications sent to the Participant.
17	Notification Detail	Notification Detail	Displays the detailed content of a notification (results, feedback, reminders, etc.).

3.1.5 Web Application Organizer Screen Flow

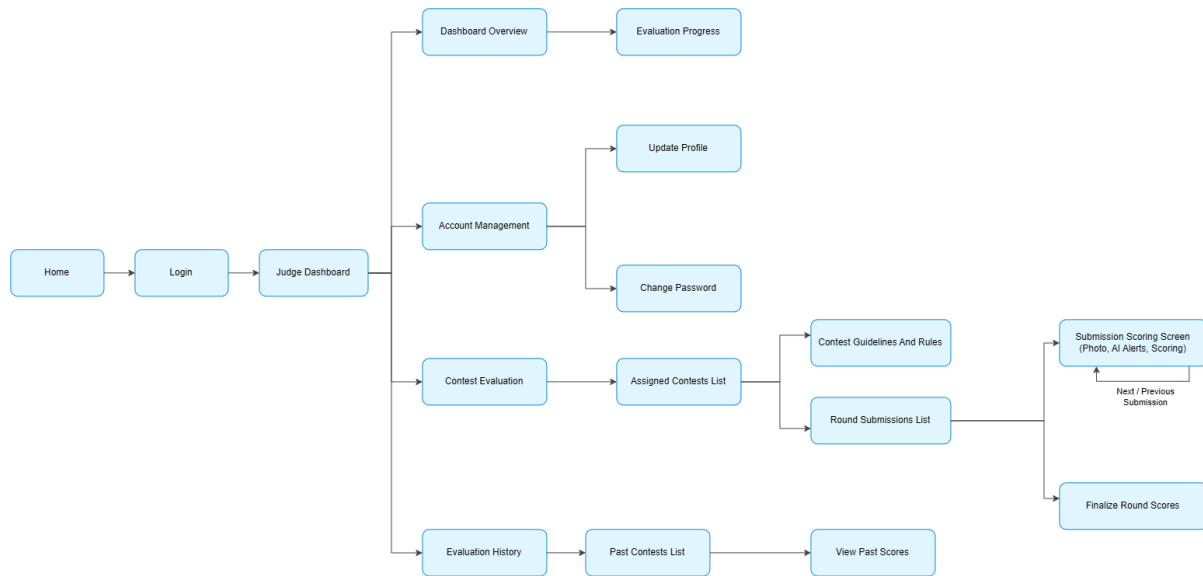


3.1.6 Web Application Organizer Screen Description

#	Feature	Screen	Description
1	Authentication	Login Page	Allows the Organizer to log in to the system using an authorized account.
2	Organizer Dashboard	Organizer Dashboard	Displays the main functions and management areas available to the Organizer.
3	Contest Setup & Configuration	Contest Setup & Configuration	Allows the Organizer to configure and manage the contest settings.
4	Contest Setup & Configuration	Enter Contest Information	Allows the Organizer to enter and update basic contest information.
5	Contest Setup & Configuration	Set Awards	Allows the Organizer to define the awards and prizes for the contest.
6	Contest Setup & Configuration	Create Criteria	Allows the Organizer to create and manage judging criteria.
7	Contest Setup & Configuration	Assign Weights	Allows the Organizer to assign weights to the judging criteria.
8	Contest Setup & Configuration	Select Judges	Allows the Organizer to select and assign judges to the contest.
9	Contest Setup & Configuration	Assign to Categories	Allows the Organizer to assign submissions to contest categories.
10	Contest Setup & Configuration	Set Schedule	Allows the Organizer to define the contest schedule and important dates.
11	Submission Management	Submission Management	Allows the Organizer to manage and process contest submissions.
12	Submission Management	Submission List	Displays the list of submissions available for management.
13	Submission Management	Flag Warnings for Organizers	Displays warnings and issues related to contest submissions.
14	Submission Management	View Flagged Submissions	Allows the Organizer to view submissions that have been flagged for review.
15	Submission Management	Request Recheck	Allows the Organizer to request a recheck of a submission when necessary.

#	Feature	Screen	Description
16	Submission Management	Reject Submission	Allows the Organizer to reject submissions that do not meet the contest requirements.
17	Submission Management	Reopen Score Sheet	Allows the Organizer to reopen a score sheet for further review or correction.
18	Results & Awards Management	Results & Awards Management	Allows the Organizer to manage scores, rankings, winners, and contest results.
19	Results & Awards Management	Calculate Total Score	Calculates the total score of submissions based on the judging criteria and weights.
20	Results & Awards Management	Rank Submissions	Displays the ranking of submissions based on their total scores.
21	Results & Awards Management	Select Winning Photos	Allows the Organizer to select the winning photographs.
22	Results & Awards Management	Publish Online Exhibition	Allows the Organizer to publish selected photographs in the online exhibition.
23	Notifications	Notifications	Allows the Organizer to manage contest-related notifications.
24	Notifications	Notify Participants	Allows the Organizer to send notifications to participants.
25	Notifications	Email & Notification Service	External service used to send emails and notifications.

3.1.7 Web Application Judge Screen Flow



3.1.8 Web Application Judge Screen Description

#	Feature	Screen	Description
1	General	Home	The landing page of the application, providing general platform information.
2	General	Login	Authentication screen allowing the Judge to access the system using secure credentials.
3	Dashboard	Judge Dashboard	The central hub routing to different modules: Dashboard Overview, Account Management, Contest Evaluation, and Evaluation History.
4	Dashboard Overview	Evaluation Progress	Displays statistical charts on pending versus completed grading tasks.
5	Account Management	Update Profile	Allows the Judge to update personal information and professional photography credentials.
6	Account Management	Change Password	Screen for updating system security credentials.
7	Contest Evaluation	Assigned Contests List	Displays a list of active film photography contests assigned to the Judge.

#	Feature	Screen	Description
8	Contest Evaluation	Contest Guidelines And Rules	Displays specific grading criteria, rules, and notes from the Organizer for the selected contest.
9	Contest Evaluation	Round Submissions List	Displays all photographic submissions within a specific round and allows navigation to evaluation or score finalization.
10	Contest Evaluation	Submission Scoring Screen	A comprehensive single-screen interface allowing the Judge to view the high-resolution photo, review AI verification alerts, input scores and comments, and navigate using Next or Previous actions.
11	Contest Evaluation	Finalize Round Scores	Summary screen accessible directly from the submissions list, enabling the Judge to review and officially submit round results to the Organizer.
12	Evaluation History	Past Contests List	Displays an archive of previously evaluated contests.
13	Evaluation History	View Past Scores	Read-only screen to review historical scores and feedback provided in past events.

3.1.9 Screen Authorization

#	Web Module / Screen	Typical Actions	Admin	Organizer	Judge	Participant
1	Authentication (Login / Register / Change PW)	C-R-U	✓	✓	✓	✓
2	Own Profile	R-U	✓	✓	✓	✓
3	Admin Dashboard	R	✓			
4	User Account & Role Management	C-R-U / Inactive	✓			
5	Organizer Dashboard	R		✓		
6	Contest Management (Create / Edit Contest)	C-R-U-D	✓	✓		
7	Round & Criteria Settings	C-R-U	✓	✓		

#	Web Module / Screen	Typical Actions	Admin	Organizer	Judge	Participant
8	Judge Assignment	C-R-U-D	✓	✓		
9	Submission Review & Moderation (Approve / Reject)	R-U-A	✓	✓		
10	AI Duplicate & AI Generated Detection Review	R-A	✓	✓		
11	Judge Grading Workspace	C-R-U			✓	
12	Contest Public Page / Explore	R	✓	✓	✓	✓
13	Submission Upload (Submit Film Photo & EXIF)	C-R-U-D				✓
14	My Submissions	R-U				✓
15	Leaderboard & Final Results	R	✓	✓	✓	✓
16	System Config & Master Data (Film Stocks, Cameras)	C-R-U-D	✓			

3.1.10 Non-screen Functions

#	Feature	System Function	Description
1	EXIF Metadata Extraction	Auto-extract Image EXIF	Automatically extract camera model, lens, aperture, shutter speed, ISO, and capture date upon photo upload. Populate extracted metadata directly into the submission record for verification.
2	AI Generated Image Detection	Async AI Detection Pipeline	Automatically scan submitted images using pre-trained AI detection models. Flag suspicious submissions exceeding the AI-probability threshold and generate analysis reports.

#	Feature	System Function	Description
3	Image Duplicate Detection	Perceptual Hashing (pHash) Scanner	Compute image feature vectors/hashes to compare with existing submissions in the database. Trigger duplicate alerts for Organizer review if similarity threshold is breached.
4	System Notification Trigger	In-app & Event-based Alerts	Send automated notifications to Participants when submission status changes (Approved/Rejected/Flagged). Notify Judges when new assignments or grading rounds are published. Notify Organizers when high-risk AI/Duplicate flags are triggered.
5	Contest Round Finalization	Leaderboard & Score Aggregator	Automatically calculate total and average scores across all criteria when a grading round closes. Rank submissions and update public Leaderboard standings.

3.1.11 Entity Relationship Diagrams

